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Fitting Segmented Regression Models by Grid Search

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Summary

A grid-search method of fitting segmented regression curves with unknown transition points is described and compared with a standard method. It is shown to be suitable for fitting a wider class of models than the standard method and to provide as a by-product a way of making reliable inferences about the abscissae of the transitions.

Keywords: MULTIPHASE; SEGMENTED REGRESSION; TRANSITION POINT

1. Introduction

We examine an alternative to the standard method of fitting segmented (sometimes called multiphase) regression models of the type

$$\begin{aligned} E(y\,|\,x) &= f_1(x, \boldsymbol{\beta}_1) \quad \text{for } X_0 \leqslant x \leqslant X_1 \\ &= f_2(x, \boldsymbol{\beta}_2) \quad \text{for } X_1 < x \leqslant X_2 \\ &= \ldots \\ &= f_r(x, \boldsymbol{\beta}_r) \quad \text{for } X_{r-1} < x \leqslant X_r \end{aligned} \tag{1}$$

with $E(y\,|\,x)$ continuous in x throughout the interval $[X_0, X_r]$, and where r, X_0, X_r and the functional form of each f_i are assumed known, but the parameters $\boldsymbol{\beta}_1, \ldots, \boldsymbol{\beta}_r$ and the vector of transition points $\mathbf{X} = (X_1, \ldots, X_{r-1})$ must be estimated from the data. Continuity of $E(y\,|\,x)$ implies that the submodels f_i are subject to constraints

$$f_i\,(X_i, \boldsymbol{\beta}_i) = f_{i+1}(X_i, \boldsymbol{\beta}_{i+1}) \tag{2}$$

for $i = 1, \ldots, r-1$. Various practical applications for such models have been found in recent years; see, for example, Sprent (1961), Hudson (1966), Dunicz (1969), Owen, Davis and Ridgman (1969), Williams (1970), Bacon and Watts (1971) and Kirby (1974).

Characteristically the data are a set of n points, $(y_1, x_1), \ldots, (y_n, x_n)$, where y_j has a variance σ^2/w_j, w_j is known but σ^2 is usually unknown *a priori.* When inferences are to be made, the y_j are assumed to be independently and normally distributed. The least-squares estimators of $\boldsymbol{\beta}_1, \ldots, \boldsymbol{\beta}_r, \mathbf{X}$ (denoted by $\hat{\boldsymbol{\beta}}_1, \ldots, \hat{\boldsymbol{\beta}}_r, \hat{\mathbf{X}}$) are the values which minimize the residual sum of squares function

$$S = \sum_{i=1}^{r} \sum_{X_{i-1} < x_j \leqslant X_i} w_j \{y_i - f_i(x_j, \boldsymbol{\beta}_i)\}^2 \tag{3}$$

subject to constraints (2). The minimum value of S is denoted by $\hat{S}$. Determining these estimates is non-trivial because S possesses potentially many local minima, and is non-differentiable at a number of points (specifically if any X_i coincides with the abscissa of an observation) which implies that some minima of S need not be stationary points. Before considering how to overcome these difficulties, we note that the fitting of segmented models which are *not* constrained to be continuous at the transition points is a distinct and easier problem : the reader is referred to the paper by Hawkins (1976) for a discussion of this.

2. Fitting the Model

The standard technique

The standard method is that described by Hudson (1966), and later refined for the "two intersecting straight lines" case by Hinkley (1969, 1971) and Singpurwalla (1974). It is a two-stage technique, the first stage being to fit the model to every feasible partition of the data without imposing continuity across the transitions. For each partition the points where adjacent f_i intersect are computed from the fitted values of $\boldsymbol{\beta}_1, ..., \boldsymbol{\beta}_r$. If these intersection points divide the data into the same partition, then the fit is admissible and its residual sum of squares noted. The second stage of the method involves one in returning to those feasible partitions which did not give admissible fits in stage one and doing the fit again, this time with constraints (2) imposed explicitly. The least-squares solution is that with smallest S overall.

Hudson's method is appropriate when the model (1) is made up entirely of linear f_i whose intersection points may be derived algebraically from $\boldsymbol{\beta}_1, ..., \boldsymbol{\beta}_r$, for in such cases it can be programmed to be fully automatic and to be relatively quick. It is not well suited to the fitting of other types of model because it requires the use of time-and-effort consuming successive approximation techniques in the fitting stage (if any f_i is non-linear), or when computing $\mathbf{X}$. Nor is it an ideal approach, whatever the model, if the user wishes to make inferences using likelihood methods (the approach generally recommended especially for transition points—Beale, 1960; Draper and Smith, 1966; Hudson, 1966; Hinkley, 1969, 1971; Feder, 1975a, b), because the necessary surfaces must be generated as a separate exercise.

The alternative fitting technique ("grid search")

We observe that a subset of the unknowns, namely the vector of transition points $\mathbf{X}$, is distinguished because S is usually more easily minimized when the value of $\mathbf{X}$ is fixed. It would appear therefore that an alternative to Hudson's method is to adapt an approach commonly used in non-linear regression and carry out a grid search over $\mathbf{X}$ to map $S_m(\mathbf{X})$, the minimum value of S at $\mathbf{X}$, to be followed by iterative refinement of the approximate minimum disclosed by the map.

This approach has two features which commend it. Firstly, it copes easily with a wider class of models than does Hudson's technique. We note that if a model is linear when $\mathbf{X}$ is fixed, $S_m(\mathbf{X})$ may be obtained algebraically by standard multiple linear regression methods. The mapping of $S_m(\mathbf{X})$, and the final refinement for such models (a far wider class than that for which Hudson's technique is suitable), is therefore very straightforward. Secondly, the method provides as a by-product the surfaces needed for making likelihood inferences about the transition abscissae. This second point is shown in the next section.

3. Inference in Segmented Regression

Because our choice of fitting method is influenced by the way inferences are to be made, I briefly review current practice. In the discussion which follows, S^0 denotes the minimum attainable value of S under hypothesis H_0, p denotes the dimension of the model's parameter space (= number of unknowns minus number of constraints), s^2 ($= \hat{S}/(n-p)$) denotes the residual mean square and $F^{\alpha}_{v,w}$ denotes the upper α per cent value of the F distribution with v and w degrees of freedom.

The LR (likelihood ratio) statistic for testing hypothesis H_0 is

$$R_1 = -2\ln(\lambda) = n\ln(S^0/\hat{S}) \tag{4}$$

which is often approximated by

$$R_2 = (S^0 - \hat{S})/s^2. \tag{5}$$

The identified case

It can be shown (Feder, 1975b) that if the model is identified (that is, if all r segments postulated in the model are known to be present and adjacent segments are distinct), and if H_0 is true, the R_1 and R_2 are asymptotically χ_q^2 (where q is the dimensional reduction in the model's parameter space imposed by H_0) and $\hat{\boldsymbol{\beta}}_1, \ldots, \hat{\boldsymbol{\beta}}_r, \hat{\mathbf{X}}$ are asymptotically normal with known variance–covariance structure.

In finite samples it is better to assume the distributions of R_1/q and R_2/q to be $F_{q,n-p}$ (Feder, 1975a). Hence, an approximate (100-α) per cent confidence region for the transition points, $\mathbf{X}$, derived from R_1 or R_2 comprises values which satisfy

$$S_m(\mathbf{X}) \leqslant C^\alpha, \tag{6}$$

where $C^\alpha = C_1^\alpha = \hat{S}\exp((r-1)F^\alpha_{r-1,n-p}/n)$ if R_1 is used, and $C^\alpha = C_2^\alpha = \hat{S} + s^2(r-1)F^\alpha_{r-1,n-p}$ if R_2 is used. Monte Carlo studies carried out by me have shown that C_2^α gives reliable results for quite small samples when all f_i are linear, and this is the value used in the examples which follow. The information needed to fix these regions is precisely the map of $S_m(\mathbf{X})$ produced by the grid-search technique.

Approximate standard errors for $\hat{\boldsymbol{\beta}}_1, \ldots, \hat{\boldsymbol{\beta}}_r$, $\hat{\mathbf{X}}$ may be obtained (Hinkley, 1971) from the information matrix conditional on the data partition implied by $\hat{\mathbf{X}}$, but without the imposition of continuity constraints (2). The partition is ambiguous if any $\hat{X}_i$ coincides with the abscissa of an observation, in which case it is reasonable to omit offending observations when computing the required information matrix. Standard errors quoted in examples are derived by this method.

The non-identified case

When fewer than r segments may actually be present (the non-identified case) the situation is more problematic. Feder (1975a) shows that if r segments are fitted when there are fewer than r present then $\hat{\boldsymbol{\beta}}_1, \ldots, \hat{\boldsymbol{\beta}}_r, \hat{\mathbf{X}}$ are not asymptotically normal, nor is the LR statistic asymptotically χ^2. The approximate methods of inference recommended for use in the identified case cannot therefore be justified with non-identified models. I do not pursue the subject further in this paper, and only identified models are treated in the examples which follow.

4. Examples

A few case studies will illustrate the grid-search technique and bring out various practical points. In the first study we look at a special case of the "two intersecting straight lines" model for which Hudson's fitting technique is unsuitable. The second study considers a model with a non-linear sub-model. Finally we look at a three-phase model, and see how to proceed with more than two segments.

Example 1. *Shoot-apex development in cereals*

Fig. 1(a) presents data from an experiment on wheat performed by Dr E. J. M. Kirby of the Plant Breeding Institute, Cambridge (Kirby, 1974). The independent variable, x, is number of days since sowing, and y is the natural logarithm of the number of primordia, a measure of shoot-apex development. Two intersecting straight lines represent the relationship between x and y quite well; that is

$$\begin{aligned} E(y\,|\,x) &= \alpha_1 + \beta_1 x \quad \text{for } x \leqslant X_1 \\ &= \alpha_2 + \beta_2 x \quad \text{for } x > X_1 \end{aligned} \tag{7}$$

subject to $\alpha_1 + \beta_1 X_1 = \alpha_2 + \beta_2 X_1$. If no other constraints are placed upon the model, it can be fitted straightforwardly by Hudson's technique. However, there are good biological grounds for believing that the transition occurs at an identifiable stage in the plant's development, namely at the end of spikelet initiation. Furthermore, an accurate estimate of the ordinate of this point is

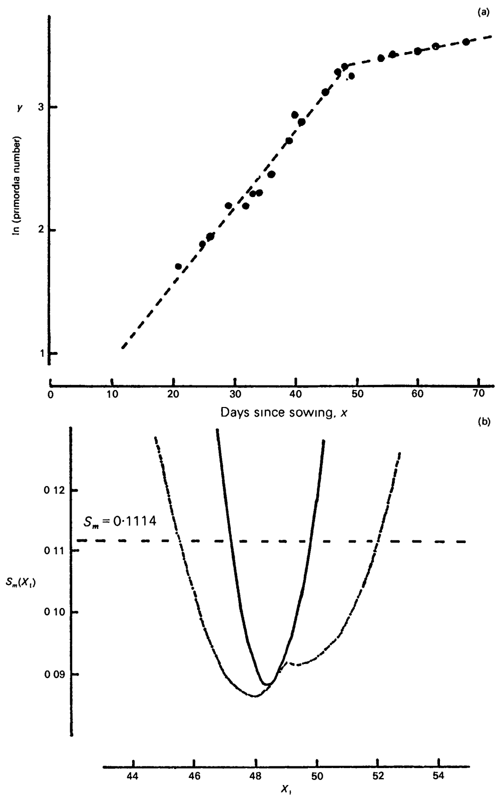

FIG. 1. (a) Data points and best-fitting line ; (b) relation between $S_m(X_1)$ and X_1 (break ordinate constrained to 3·3358 shown —, break ordinate unconstrained shown -·-·-) for Example 1.

available from mature plants : for these data it was 3·3358. It is desirable to be able to use this information when fitting the model in order to test whether the true ordinate of the break-point differs from the specified value and, if it does not, to improve the precision with which the remaining unknown parameters are estimated.

There is no simple way in which Hudson's technique can be applied to this more restricted model. The grid-search approach, on the other hand, copes quite easily because, for fixed X_1, the model is linear. The results are shown in Fig. 1(a), where the fitted curve is superimposed on the data points, and in Fig. 1(b) where $S_m(X_1)$ is plotted against X_1 in the range $44 \leqslant X_1 \leqslant 54$. The function $S_m(X_1)$ was evaluated at intervals of 0·25 along the X_1-axis, and individual points joined up, to give a continuous curve. The minimal value $\hat{S}$ ($= 0{\cdot}08828$) was located by linear search at $\hat{X}_1' = 48{\cdot}4$.

From (6) an approximate 95 per cent confidence region for X_1 is given by

$$S_m(X_1) \leqslant 0{\cdot}1114.$$

A horizontal line at this height in Fig. 1(b) indicates this region to be $47{\cdot}2 \leqslant X_1 \leqslant 49{\cdot}8$.

Fig. 1(b) also shows the plot of $S_m(X_1)$ when the model is not constrained to have a specified ordinate at X_1. The minimal value $\hat{S}$ in this case was 0·08645, and was located at $\hat{X}_1 = 48{\cdot}0$ which is the abscissa of an observation, and not therefore a stationary point of S. The LR test using R_2 showed no significant evidence that the break ordinate differs from the specified value ($R_2 = 0{\cdot}34$ with 1 and 16 degrees of freedom). The 95 per cent confidence region for X_1 when the model is not constrained to have a specified ordinate at the break is given by $S_m(X_1) \leqslant 0{\cdot}1107$, that is the interval $45{\cdot}6 \leqslant X_1 \leqslant 52{\cdot}0$; this interval is almost two and a half times as wide as that obtained when the break ordinate is specified and shows the usefulness of being able to impose the value of that ordinate.

Example 2. Two phases, one of them non-linear

We consider an arbitrarily chosen model

$$\begin{aligned} E(y \mid x) &= \alpha + \beta \cos(x - \gamma) \quad \text{for } x \leqslant X_1 \\ &= \alpha + \beta \cos(X_1 - \gamma) \quad \text{for } x > X_1, \end{aligned} \tag{8}$$

and observe that the requirement of continuity at the transition is incorporated into the model (see Section 5 below for discussion of constraint handling with non-linear f_i). Twenty x-values were sampled from a uniform distribution in the range $0 < x < 1$, and the corresponding y-values formed from the addition to (8) (with parameter values $\alpha = 0$, $\beta = 1$, $\gamma = 0{\cdot}7$ and $X_1 = 0{\cdot}5$) of error terms sampled from a normal distribution with standard deviation 0·02. The resulting data points are shown in Fig. 2(a).

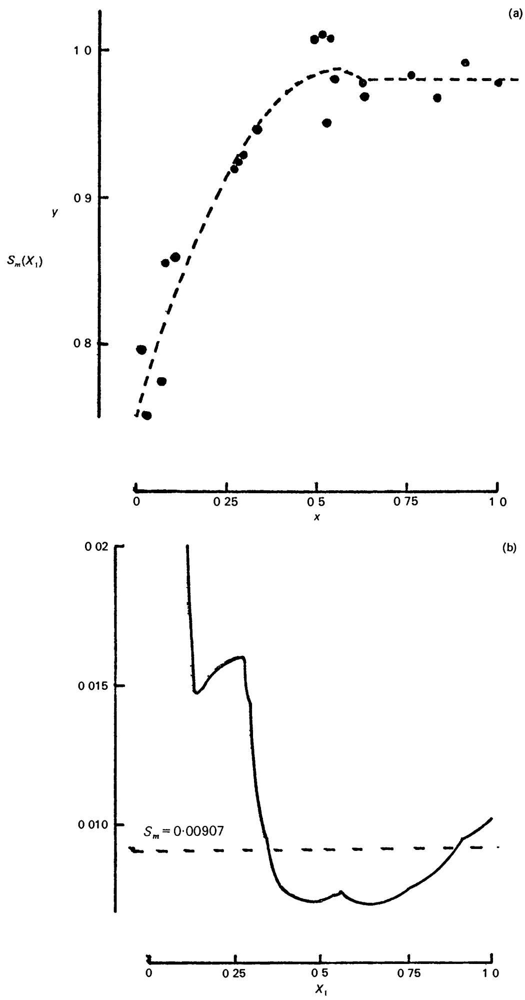

FIG. 2. (a) Data points and best-fitting line; (b) relation between $S_m(X_1)$ and X_1, for Example 2.

The function $S_m(X_1)$ was evaluated at intervals of 0·02 in the range $0{\cdot}07 \leqslant X_1 \leqslant 0{\cdot}99$ (the entire range of x-values in the data) using a Marquardt technique (Draper and Smith, 1966; Adby and Dempster, 1974). Final values of α, β and γ from one minimization were used as starting values in the next; a strategy which makes minimization rapid and reliable. The results are shown in Fig. 2(b). The minimal value $\hat{S}$ (= 0·00710) was located at $\hat{X}_1 = 0{\cdot}63$. The fitted curve is shown superimposed on the data in Fig. 2(a). An approximate 95 per cent confidence region for X_1 using criterion (6) satisfies $S_m(X_1) \leqslant 0{\cdot}00907$ which Fig. 2(b) shows to be the interval $0{\cdot}34 \leqslant X_1 \leqslant 0{\cdot}90$.

The shape of the relationship in Fig. 2(b) deserves some comment. Of particular interest are the turning points at $X_1 = 0{\cdot}14$ and $X_1 = 0{\cdot}26$, both of which are interior points of the same interval between adjacent observations. Such behaviour is theoretically possible, but rare in practice. Secondly, the existence of two minima with almost equal values of S_m, one at $X_1 = 0{\cdot}63$ and the other at $X_1 = 0{\cdot}48$, demonstrates how useful a plot of $S_m(\mathbf{X})$ can be. Although $X_1 = 0{\cdot}63$ is the best estimate, $X_1 = 0{\cdot}48$ is almost as good. A fair description of the results would require this fact to be mentioned. Without a plot of $S_m(X_1)$ it might have escaped notice. Finally, the curve, with its three minima, shows that a confidence region need not be a single connected set: it could be a collection of disjoint sets.

Example 3. *Voluntary feed intake in ruminants*

One of the models proposed for relating voluntary feed intake in ruminants, y, to metabolizable energy (ME) concentration in the diet, $z = 1/x$, is given by Owen *et al.* (1969) as

$$\begin{aligned} E(y\,|\,x) &= \alpha x && \text{for } x \leqslant X_1 \\ &= \beta && \text{for } X_1 < x \leqslant X_2 \\ &= \gamma x/(x-1) && \text{for } X_2 < x \end{aligned} \tag{9}$$

subject to continuity requirements $\alpha X_1 = \beta = \gamma X_2/(X_2 - 1)$. So far it has not been possible to produce experimental data in which all three segments of the model are adequately represented,

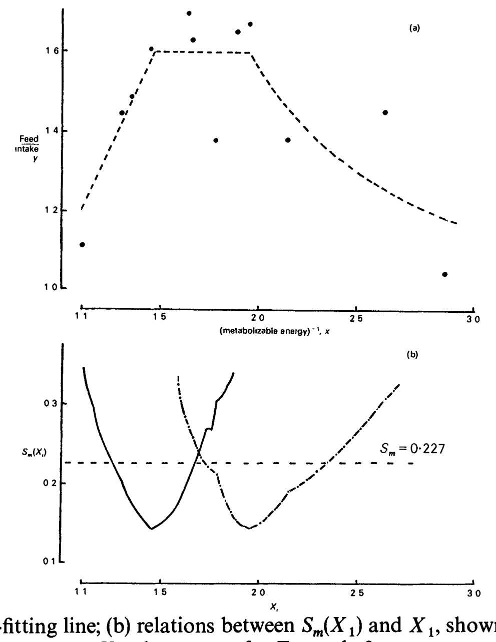

FIG. 3. (a) Data points and best-fitting line; (b) relations between $S_m(X_1)$ and X_1, shown —, and between $S_m(X_2)$ and X_2, shown -----, for Example 3.

and for the purpose of this example I use artificial data (Fig. 3(a)) similar to those given by Owen *et al.* (1969). Twelve ME values were chosen, and the corresponding y values were derived from the addition to (9) (with parameter values $\alpha = 1$, $\beta = 1{\cdot}5$, $\gamma = 0{\cdot}75$, $X_1 = 1{\cdot}5$, $X_2 = 2$) of error terms sampled from a normal distribution with standard deviation 0·1.

The function $S_m(X_1, X_2)$ was calculated at intervals of 0·05 along both the X_1 and X_2 axes in the region $1{\cdot}1 \leqslant X_1 \leqslant X_2 \leqslant 2{\cdot}95$, the entire range of x in the data. The resulting contour map is shown in Fig. 4. There is evidently only one minimum ($\hat{S} = 0{\cdot}145$) which is located at $(\hat{X}_1, \hat{X}_2) = (1{\cdot}46, 1{\cdot}95)$. An approximate 95 per cent confidence region for (X_1, X_2) using criterion (6) is bounded by the isobar $S_m(\mathbf{X}) = 0{\cdot}282$ and is indicated in Fig. 4 as contour 2. The best fitting

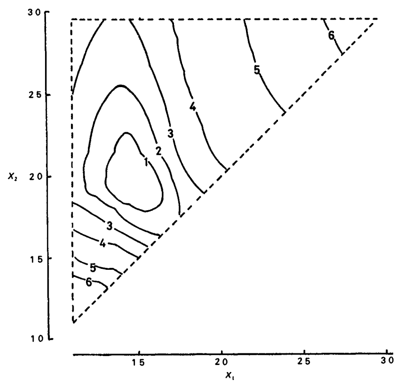

FIG. 4. Contours of $S_m(X_1, X_2)$ for Example 3. Values of S_m defining the contours are: (1) 0·208, (2) 0·282, (3) 0·403, (4) 0·673, (5) 1·5, (6) 2·5. Contours 1, 2, 3 and 4 are the boundaries of an 80, 95, 99 and 99·9 per cent confidence region, respectively, for (X_1, X_2).

curve is shown in Fig. 3(a). Fig. 3(b) displays plots of approximations to $S_m(X_1)$ and $S_m(X_2)$, the minimum values of S for fixed X_1 and X_2 respectively. These approximations were obtained from the map of $S_m(X_1, X_2)$, and from Fig. 3(b) the 95 per cent confidence regions for X_1 and X_2 separately are obtained as those values satisfying $S_m(X_i) \leqslant 0{\cdot}227$ $(i = 1, 2)$, namely $1{\cdot}25 \leqslant X_1 \leqslant 1{\cdot}68$ and $1{\cdot}72 \leqslant X_2 \leqslant 2{\cdot}35$

5. DISCUSSION

Calculating $S_m(\mathbf{X})$

If all f_i in the model (1) are linear, an efficient way of performing the necessary calculations is first to fit the model without imposing constraints (2), and then to use Lagrangian methods to obtain the adjustments to the residual sum of squares, and to the values of $\boldsymbol{\beta}_1, \ldots, \boldsymbol{\beta}_r$, when continuity at a specific $\mathbf{X}$ is imposed. The method is efficient because the same unconstrained fit will apply to a number of $\mathbf{X}$-values on the grid. Thus, it need only be carried out once for those $\mathbf{X}$, and several values of $S_m(\mathbf{X})$ may be obtained by the relatively quick constraining process. The necessary calculations are as follows. If the least-squares parameter estimates and residual sum of squares for a linear model $\mathbf{y} = X\boldsymbol{\beta} + \mathbf{e}$ are denoted by $\tilde{\boldsymbol{\beta}}$ and $\tilde{S}$, then the adjusted values when constraints $H\boldsymbol{\beta} = \mathbf{c}$ are imposed are (Plackett, 1960, p. 53)

$$\begin{aligned} \boldsymbol{\beta} &= \tilde{\boldsymbol{\beta}} - (X'WX)^{-1} H'[H(X'WX)^{-1} H']^{-1}(H\tilde{\boldsymbol{\beta}} - \mathbf{c}), \\ S &= \tilde{S} + (H\tilde{\boldsymbol{\beta}} - \mathbf{c})'[H(X'WX)^{-1} H']^{-1}(H\tilde{\boldsymbol{\beta}} - \mathbf{c}), \end{aligned} \tag{10}$$

where W is the diagonal matrix of weights ($W_{jj} = w_j$).

If any of the f_i are non-linear, even when $\mathbf{X}$ is fixed, the value of $S_m(\mathbf{X})$ will usually need to be determined by iterative minimization methods. In such cases the usual problems associated with non-linear regression will be encountered. Ross (1970) gives a good discussion of the efficient use of iterative methods in non-linear least squares.

The imposition of continuity constraints with non-linear f_i can be handled in a variety of ways (see Adby and Dempster, 1974, Chapter 5). A simple method is to reduce the problem to an unconstrained minimization by incorporating continuity directly into the model (see Example 2 above), but this is not always possible.

Displaying $S_m(\mathbf{X})$

The shape of $S_m(\mathbf{X})$ is easily apprehended if it is displayed graphically rather than in tabular form. This presents no difficulties if the number of transition points is one or two; $S_m(X_1)$ can be plotted against X_1 if there is only one transition (Figs. 1(b), 2(b)), and if there are two, a contour map of $S_m(X_1, X_2)$ can be produced (Fig. 4). For three or more breaks it is simplest to display lower-dimensional extracts of $S_m(\mathbf{X})$. A sensible choice is $S_m(\mathbf{V})$, the minimum value of S for the fixed value of a subvector, $\mathbf{V}$, of transition points (Fig. 3(b)), since it is the function required for deriving confidence regions for $\mathbf{V}$. The value of $S_m(\mathbf{V})$ can be approximated from the map of $S_m(\mathbf{X})$ provided the grid is suitably fine and spans the required minimum.

Final refinement of $\hat{\mathbf{X}}$

Since $\hat{S}$ may not be a stationary point of S, its location from the approximation revealed by the map is best achieved by a technique which does not employ derivatives. Linear search (Adby and Dempster, 1974) is suitable if there is only one transition point, and pattern search (Powell, 1964; Adby and Dempster, 1974) for two or more breaks.

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